Functional Requirements Document (FRD)

Merchant Intelligence Platform (MIP) for Cards Business

Version: 1.0
Project: Merchant Intelligence Platform (MIP)
Prepared From: BRD v2.0
Document Type: Functional Requirements Document (FRD)

---

1. Purpose

The Merchant Intelligence Platform (MIP) will provide a centralized analytical platform that consolidates merchant, transaction, customer, acquiring, geographic, loyalty, digital engagement, affinity, and competitive intelligence data into a unified merchant intelligence ecosystem.

The platform will enable:

- Merchant 360 Intelligence
- Merchant Segmentation
- Merchant Performance Monitoring
- AI-driven Recommendations
- Merchant Growth Prediction
- Cross-Sell Identification
- Churn Detection
- Portfolio Performance Management

---

2. Functional Scope

Module	Description
MIP-01	Merchant 360 Intelligence
MIP-02	Merchant Segmentation
MIP-03	Merchant Performance Dashboard
MIP-04	KPI Governance Layer

Module	Description
MIP-05	Alert & Notification Engine
MIP-06	AI Insights & Recommendations
MIP-07	Reporting & Export Services
MIP-08	Security & Access Management

---

3. User Roles

Business Users

Role	Access Level
Cards Business Head	Portfolio-wide access
Merchant Acquiring Team	Merchant & Segment access
Relationship Managers	Assigned Merchant access
Product Team	Merchant & Product Analytics
Marketing Team	Campaign & Segmentation access
Finance Team	Revenue/KPI Validation
Analytics Team	Full Analytical Access

---

4. System Architecture Overview

Data Sources

- Card Transaction Systems
- Merchant Master System
- Acquiring Platform
- Customer Data Platform
- Merchant Lending Systems
- Loyalty System

- Digital Engagement Sources
- Competitive Intelligence Sources

Data Flow

Source Systems ↓ Raw Data Landing Zone ↓ Data Quality Validation ↓ Enterprise Data Warehouse ↓ Merchant Intelligence Data Mart ↓ KPI Engine ↓ Dashboards / Alerts / Reports

---

5. Functional Requirements

BR-01 Merchant 360 Intelligence

FR-01 Merchant Search

Description

Users shall search merchants using:

- Merchant ID
- Merchant Name
- MCC Code
- Category
- Relationship Manager
- City
- State
- Acquiring Status

Priority

Critical

Acceptance Criteria

- Search response < 3 seconds
- Partial text search supported
- Auto-suggestion supported

FR-02 Merchant Profile Dashboard

Description

System shall display complete Merchant 360 Profile.

Display Sections

Merchant Information

- Merchant ID
- Merchant Name
- Business Type
- Merchant Category
- MCC
- Annual Revenue
- Merchant Size
- Relationship Manager
- Onboarding Date

Geographic Information

- City
- State
- Latitude
- Longitude
- Catchment Area

Acquiring Information

- POS Count
- MDR Revenue
- Settlement Volume
- Acquirer Information

Credit Information

- Credit Score
- Eligible Lending Limit
- Existing Loans

- Cash Flow Stability

Digital Maturity Information

- Website Presence
- Mobile App Presence
- E-commerce Availability
- Digital Payment Adoption

FR-03 Revenue Growth KPI

Description

System shall calculate Merchant Revenue Growth.

Formula

Revenue Growth %

=

(Current Period Revenue − Previous Period Revenue)

/

Previous Period Revenue

×100

Display Locations

- KPI Tile
- Merchant Scorecard
- Trend Chart

Drill Down

Revenue KPI → Monthly Revenue → Transaction Details

FR-04 Wallet Share KPI

Formula

Wallet Share %

=

Bank Merchant Volume

/

Total Merchant Market Volume

×100

Functional Requirements

System shall:

- Calculate wallet share monthly
- Compare against peers
- Highlight growth opportunities

FR-05 Merchant Attrition KPI

Description

Identify merchants becoming inactive.

Business Rules

Merchant classified as Lost when:

- No transaction activity for 90 days OR
- Merchant acquiring relationship terminated

Thresholds

Risk Level	Condition
Low	< 10% decline
Medium	10%-25% decline
High	>25% decline

FR-06 Customer Analytics Panel

Features

Display:

- Customer Count
- Active Customers
- Repeat Customers
- Income Segments
- Category Preferences
- Spend Behavior

Visualizations

- Pie Charts
- Heat Maps
- Trend Charts

FR-07 Merchant Affinity Network

Description

Display merchant-to-merchant relationships based on shared customers.

Functional Capabilities

- Network Graph
- Affinity Score Calculation
- Shared Customer Count
- Top Partner Recommendations

FR-08 Cross-Sell Recommendation Engine

Description

Recommend banking products.

Products

- Merchant Loans
- POS Upgrade
- Working Capital

- Business Credit Cards
- Trade Services
- Cash Management

Output

Merchant	Recommendation	Score	Expected Revenue

FR-09 Merchant Risk Dashboard

Display

- Churn Score
- Credit Score
- Revenue Decline Score
- Lending Eligibility
- Risk Category

BR-02 Merchant Segmentation

FR-10 Automated Segmentation Engine

Description

System shall automatically classify merchants.

Revenue Segmentation

Segment	Revenue Criteria
Platinum	Top Revenue Band
Gold	High Revenue Band
Silver	Medium Revenue Band
Bronze	Low Revenue Band

Business Size Segmentation

Segment
Enterprise
Mid-Market
SMI
Micro

Growth Stage Segmentation

Segment
Emerging
Growth
Mature
Declining

FR-11 Segment Migration Tracking

Description

Track movement between segments.

Example

Bronze → Silver

Silver → Gold

Growth → Mature

Mature → Declining

Outputs

- Heatmaps
- Migration Matrix
- Segment Trends

FR-12 Active Merchant Count KPI

Formula

Count Distinct Active Merchant IDs

Views

- Portfolio
- Segment
- Geography
- Category

FR-13 Segment Performance Dashboard

Display

KPIs

- Active Merchant Count
- Revenue Growth %
- Attrition Rate

Tables

- Segment Revenue
- Segment Profitability
- Merchant Distribution

FR-14 Churn Risk Segment Detection

Description

Identify segments with:

- Declining Growth
- High Attrition
- Reduced Wallet Share

Outputs

- Alert Generation
- RM Follow-up List

FR-15 Segment Opportunity Finder

Description

System identifies:

- High Growth Segments
- High Wallet Share Potential Segments
- Cross-Sell Potential Segments

BR-03 Merchant Performance Dashboard

FR-16 Executive KPI Dashboard

Display Headline KPIs

KPI 1

Total Sales

SUM(Transaction Amount)

KPI 2

Transaction Volume

COUNT(Transaction)

KPI 3

Merchant Wallet Share

Bank Merchant Volume

/

Market Merchant Volume

``

FR-17 Merchant Growth Trend Analysis

Features

- Daily Trend

- Weekly Trend
- Monthly Trend
- Quarterly Trend
- Yearly Trend

Comparison

- MoM
- QoQ
- YoY

FR-18 Top Category Analysis

Display

Top 10 Categories by:

- Sales
- Transactions
- Revenue Growth

FR-19 Geographic Performance Dashboard

Features

- State Performance
- City Performance
- PIN Code Analysis
- Merchant Density Analysis

Visuals

- Geo Maps
- Heat Maps
- Ranking Tables

FR-20 Merchant Performance Ranking

Ranking Metrics

- Total Sales
- Wallet Share
- Revenue Growth
- Transaction Growth

Output

Top 10 Merchants

Bottom 10 Merchants

FR-21 Performance Alerts

Trigger Conditions

Alert Type	Threshold
Sales Drop	>20%
Transaction Drop	>15%
Attrition Risk	High
Wallet Share Decline	>10%

Delivery Channels

- Dashboard
- Email
- Notification Center

---

6. AI Insight Requirements

FR-22 Growth Prediction

System shall forecast:

- Merchant Revenue
- Merchant Sales
- Transaction Volume

Forecast Horizons:

- 30 Days
- 90 Days
- 180 Days

FR-23 Churn Prediction

Inputs

- Sales Trend
- Transaction Trend
- Wallet Share Trend
- Merchant Engagement

Outputs

- Churn Probability
- Risk Category
- Recommended Action

FR-24 Next Best Action Recommendation

System shall provide recommendations such as:

- Offer Merchant Loan
- Offer POS Upgrade
- Retention Campaign
- Loyalty Enrollment
- Cross-Sell Products

---

7. Reports Requirements

Standard Reports

Merchant Portfolio Report

Frequency:

- Daily
- Weekly
- Monthly

Merchant Revenue Report

Frequency:

- Monthly

Merchant Attrition Report

Frequency:

- Weekly

Wallet Share Report

Frequency:

- Monthly

Segmentation Migration Report

Frequency:

- Monthly

Cross-Sell Opportunity Report

Frequency:

- Weekly

---

8. Data Requirements

Fact Tables

Fact_Transactions

Field
Transaction_ID
Merchant_ID
Customer_ID
Card_ID

Field
Transaction_Date
Transaction_Amount
Channel
MCC

Fact_Merchant_Revenue

Field
Merchant_ID
Revenue_Date
Revenue_Amount

Fact_Customer_Spend

Fact_Acquiring_Revenue

Fact_Merchant_Affinity

Fact_Campaigns

Dimension Tables

- Dim_Merchant
- Dim_Time
- Dim_Customer
- Dim_Card
- Dim_Channel
- Dim_Location
- Dim_Merchant_Category
- Dim_Partner

---

9. Security Requirements

Role Based Access Control

Role	Merchant Data	Customer Data
RM	Assigned Only	Masked
Business Head	All	Masked
Analyst	All	Tokenized
Admin	All	Tokenized

Mandatory Controls

- Tokenization
- Data Masking
- Encryption At Rest
- Encryption In Transit
- Audit Logging
- MFA Authentication

---

10. Non-Functional Requirements

Requirement	Target
Dashboard Load Time	<5 Sec
Search Response	<3 Sec
Data Refresh SLA	Daily
Platform Availability	99.9%
Concurrent Users	500+
KPI Accuracy	>99.5%
Export Generation	<30 Sec

---

11. BRD to FRD Traceability Matrix

BR ID	Business Requirement	FR Mapping
BR-01	Merchant 360 Intelligence	FR-01 to FR-09
BR-02	Merchant Segmentation	FR-10 to FR-15
BR-03	Merchant Performance Dashboard	FR-16 to FR-21
AI Insights	Predictive Intelligence	FR-22 to FR-24
Reporting	KPI & Operational Reporting	Report Requirements
Security	Data Protection	Security Requirements

---

12. UAT Acceptance Criteria

Merchant 360

- Merchant profile loads successfully.
- KPI calculations match approved formulas.
- Drill-through works to transaction level.

Segmentation

- Merchant assigned to correct segment.
- Segment migrations calculated correctly.

Performance Dashboard

- KPI values reconcile with source systems.
- Geography and category reports display accurately.

AI Insights

- Churn scores generated successfully.
- Recommendations generated for eligible merchants.

Security

- Customer data masked/tokenized.
- RBAC enforced successfully.

End of Functional Requirements Document (FRD).